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STA2002 Probability and Statistics II **Spring 2023**

Supplementary 1*: Asymptotic Distributions of Sample Variance

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This lecture notes help introduce some asymptotic properties of the **sample variance** of a general **random sample** which can be normal distributed or non-normal distributed. Recall that if a random sample is normal, we have Student's Theorem which has been introduced in lecture slides 2.

1.1 Preliminary

Let a sample of size n of i.i.d. random variables X_i, $i = 1, \ldots, n$, with mean μ and variance σ^2. Set the sample mean and the sample variance as

$$\overline{X} := \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad S^2 := \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2 \tag{1}$$

Lemma 1.1.1. *The expectation and variance of sample variance S^2 are,*

$$\mathbb{E}(S^2) = \sigma^2, \qquad Var(S^2) = \frac{1}{n}\left(\mu_4 - \frac{n-3}{n-1}\sigma^4\right),$$

where $\mu_4 = \mathbb{E}(X_i - \mu)^4$.

Theorem 1.1.2 (Consistency of sample mean)**.** *Sample mean $\overline{X}$ is a consistent estimator of μ,*

$$\overline{X} \xrightarrow{P} \mu.$$

Theorem 1.1.3 (Consistency of sample variance)**.** *Sample variance S^2 is a consistent estimator of σ^2,*

$$S^2 \xrightarrow{P} \sigma^2.$$

Proof. First manipulate the expression of S^2 as follows,

$$\begin{aligned} S^2 &= \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2 \\ &= \frac{n}{n-1}\left(\frac{1}{n}\sum_{i=}^{n} X_i^2 - \overline{X}^2\right) \end{aligned}$$

By Theorem 1.1.2,

$$\frac{1}{n}\sum_{i=}^{n} X_i^2 \xrightarrow{P} \mathbb{E}(X_1^2), \quad \text{as } n \to \infty.$$

By Theorem 1.1.2 and property of " $\xrightarrow{P}$ ",

$$\overline{X}^2 \xrightarrow{P} \mu^2, \quad \text{as } n \to \infty.$$

Meanwhile,

$$\frac{n}{n-1} \longrightarrow 1, \quad \text{as } n \to \infty.$$

Thus, we have that,

$$S^2 \xrightarrow{P} 1 \cdot [\mathbb{E}(X_1^2) - \mu^2] = \sigma^2, \quad \text{as } n \to \infty.$$

□

Theorem 1.1.4 (Slutsky's Theorem)**.** *Let X_n, X, A_n, and B_n be random variables and let a and b be constants. If $X_n \xrightarrow{D} X$, $A_n \xrightarrow{P} a$, and $B_n \xrightarrow{P} b$, then*

$$A_n + B_n X_n \xrightarrow{D} a + bX, \quad \textit{as } n \to \infty.$$

1.2 More Asymptotic Properties

Theorem 1.2.1 (Central Limit Theorem)**.** *Suppose that $X_1, X_2, \ldots, X_n$ are i.i.d. with common mean μ and finite variance σ^2. Then the limiting distribution of*

$$W = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \frac{\sum_{i=1}^{n} X_i - n\mu}{\sqrt{n}\sigma}$$

is standard normal distribution $N(0,1)$. Equivalently,

$$\sqrt{n}(\overline{X} - \mu) \xrightarrow{D} N(0,\ \sigma^2).$$

Theorem 1.2.2. *The limiting distribution of* $\sqrt{n}(S^2 - \sigma^2)$ *is normal distribution with mean zero and variance* $\mu_4 - \sigma^2$,

$$\sqrt{n}(S^2 - \sigma^2) \xrightarrow{D} N(0,\ \mu_4 - \sigma^4).$$

Proof. First manipulate the expression of S^2 as follows,

$$\begin{aligned}(n-1)S^2 &= \sum_{i=1}^{n} \left[(X_i - \mu) - (\overline{X} - \mu)\right]^2 \\ &= \sum_{i=1}^{n} (X_i - \mu)^2 - n(\overline{X} - \mu)^2.\end{aligned}$$

Therefore,

$$\sqrt{n}(S^2 - \sigma^2) = \frac{\sqrt{n}}{n-1} \sum_{i=1}^{n} (X_i - \mu)^2 - \frac{n\sqrt{n}}{n-1}(\overline{X} - \mu)^2 - \sqrt{n}\sigma^2.$$

Next, manipulate the expression of $\sqrt{n}(S^2 - \sigma^2)$ further,

$$\begin{aligned}\sqrt{n}(S^2 - \sigma^2) &= \frac{\sqrt{n}}{n-1} \sum_{i=1}^{n} (X_i - \mu)^2 - \frac{n\sqrt{n}}{n-1}(\overline{X} - \mu)^2 - \sqrt{n}\sigma^2 \\ &= \frac{n\sqrt{n}}{n-1} \frac{1}{n} \sum_{i=1}^{n} (X_i - \mu)^2 - \frac{\sqrt{n}(n-1)}{n-1}\sigma^2 - \frac{n\sqrt{n}}{n-1}(\overline{X} - \mu)^2 \\ &= \frac{n\sqrt{n}}{n-1} \left[\frac{1}{n} \sum_{i=1}^{n} (X_i - \mu)^2 - \sigma^2\right] + \frac{\sqrt{n}}{n-1}\sigma^2 - \frac{n\sqrt{n}}{n-1}(\overline{X} - \mu)^2,\end{aligned}$$

and analyze the limiting behavior of each term in the last expression, respectively.

(1) For the 2nd term $\frac{\sqrt{n}}{n-1}\sigma^2$ which is real number,

$$\frac{\sqrt{n}}{n-1}\sigma^2 \ \to\ 0, \quad \text{as } n \to \infty.$$

(2) For the 3rd term $\frac{n\sqrt{n}}{n-1}(\overline{X} - \mu)^2$,

$$\frac{n\sqrt{n}}{n-1}(\overline{X} - \mu)^2 = \frac{n}{n-1} \cdot \sqrt{n}(\overline{X} - \mu) \cdot (\overline{X} - \mu)$$

where,

$$\frac{n}{n-1} \ \to\ 1, \quad \text{as } n \to \infty,$$

and by Central Limit Theorem,

$$\sqrt{n}(\overline{X} - \mu) \xrightarrow{D} N(0, \sigma^2), \quad \text{as } n \to \infty,$$

and by Theorem 1.1.2

$$(\overline{X} - \mu) \xrightarrow{P} 0, \quad \text{as } n \to \infty.$$

Then by Slutsky's Theorem the following product converges in probability to zero,

$$\sqrt{n}(\overline{X} - \mu)^2 \xrightarrow{P} 0, \quad \text{as } n \to \infty,$$

thus, the 3rd term converges in probability to zero.

(3) For the 1st term,

$$\frac{n\sqrt{n}}{n-1}\left[\frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2 - \sigma^2\right] = \frac{n}{n-1} \cdot \sqrt{n}\left[\frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2 - \sigma^2\right],$$

where $\frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2$ is the sample mean of $(X_1 - \mu)^2$, and σ^2 is the expectation of $(X_1 - \mu)^2$. Further more,

$$\text{Var}\left[(X_1 - \mu)^2\right] = \mu_4 - \sigma^4.$$

Thus by Central Limit Theorem,

$$\sqrt{n}\left[\frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2 - \sigma^2\right] \xrightarrow{D} N(0,\ \mu_4 - \sigma^4), \quad \text{as } n \to \infty.$$

Together with,

$$\frac{n}{n-1} \to 1, \quad \text{as } n \to \infty,$$

yields that the 1st term converges in distribution to the normal distribution $N(0,\ \mu_4 - \sigma^4)$.

In summary, $\sqrt{n}(S^2 - \sigma^2)$ converges in distribution to the normal distribution $N(0,\ \mu_4 - \sigma^4)$. □

1.3 Further Reading

For the formal mathematical definition and discussion of the concepts **convergence in probability** " $\xrightarrow{P}$ " and **convergence in distribution** " $\xrightarrow{D}$ ", please read Section 5.1 and Section 5.2 of the recommended reading book [1].

References

[1] Hogg, R. V., McKean, J., Craig, A. T. (2018). Introduction to mathematical statistics. Pearson Education.